

Gambling with Graduation: Mobile Sports Betting Legalization and Student Persistence

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Abstract

This paper examines whether the legalization of mobile sports betting affects college student persistence. I construct an original dataset identifying the precise date legal mobile sports betting came online in each state, distinct from the date of legislative authorization, and combine it with an annual panel of 1,488 four-year colleges from the Integrated Postsecondary Education Data System (IPEDS), 2014–2024. Using a difference-in-differences design that accounts for staggered treatment timing, I find that legalization is associated with a statistically significant 1.21 percentage point decline in four-year graduation rates, with effects increasing in cohorts’ cumulative exposure to legal betting access. I also show that the gap between female and male six-year graduation rates grows by 0.6 percentage points. The effect is absent from both overall six-year graduation rates and first-year retention, the latter serving as a falsification check since most entering freshmen are below the legal betting age of 21. Checks on applicant, admissions, and enrollment composition show no evidence that the results are driven by selection into treated institutions.

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1 Introduction

Since the Supreme Court’s 2018 ruling in *Murphy v. NCAA*, legal sports betting has grown into a major industry. Legal sportsbooks in the United States took in nearly \$148 billion in bets from Americans in 2024, with 95 percent of that handle wagered online rather than at a physical sportsbook ([American Gaming Association, 2025](#)). For most Americans today, placing a legal bet on a sporting event requires nothing more than a smartphone and an internet connection.

This growth in mobile sports betting access has led to uneven uptake across the population: survey evidence finds that sports bettors are disproportionately young and male, with nearly half of men aged 18 to 49 reporting an active online sports betting account ([Siena College Research Institute, 2025](#)). A new literature has begun to trace how sports betting legalization reshapes daily behavior alongside spending. Using multiple waves of the American Time Use Survey (ATUS), [Fleming et al. \(2024\)](#) find that mobile sports betting legalization increased time spent on leisure activities nationally, driven by increases in time spent consuming mass media and socializing, and that this reallocation of time was more pronounced among young men. Taken together, this evidence suggests that both the financial and behavioral consequences of mobile sports betting access are concentrated in exactly the demographic that populates American college campuses, yet no existing research asks whether this reallocation of time and money extends into educational outcomes.

This paper studies whether this rapid expansion of mobile sports betting access has affected outcomes for college students, complementing a growing literature on the financial and behavioral effects of sports betting legalization with evidence on educational attainment. College students are a uniquely relevant population to study for two further reasons. Mobile access places a sportsbook within reach at any time, including during class, in a dorm room, or between assignments, lowering the cost of betting precisely for a population with substantial unstructured time. College students are also unusually close to the object of wagering, since their own institutions field athletic teams on which bets can be placed. A 2023 NCAA survey found that 67 percent of students living on campus report betting on sports, and that 41 percent of student bettors have wagered on their own school’s teams ([NCAA, 2023](#)), underscoring that this proximity to wagering extends to the broader student body as fans and bettors, not just athletes themselves.

Concern about gambling on campus also predates mobile legalization: prior research has documented gambling as an emerging health issue among students, found sports betting to be common among athletes, fans, and students more broadly, characterized bettors as disproportionately young and male, and examined the role of perceived skill relative to luck in motivating participation (Stuhldreher et al., 2007; Nelson et al., 2007; Humphreys and Pérez, 2012; Kim and López, 2025).

I construct an original dataset identifying the date on which each state first offered legal mobile sports betting to the public, and combine it with an annual panel of 1,488 four-year colleges drawn from the Integrated Post-secondary Education Data System (IPEDS), spanning 2014 to 2024. Using this panel, I examine whether the timing of a state’s legalization of mobile sports betting affected graduation rates at colleges within that state.

I identify the impact of legalized mobile sports betting on graduation rates using a difference-in-differences design comparing colleges in never-treated states to colleges in states legalizing mobile betting, exploiting the staggered timing of state-level legalization across the sample period. Because standard two-way fixed effects estimators can be biased in staggered settings, I implement the estimator developed by Callaway and Sant’Anna (2021) alongside the standard approach. I also examine first-year retention rate as a falsification check: because nearly every state restricts sports betting to individuals aged 21 or older, entering freshmen are largely unable to legally wager, so a true behavioral channel operating through betting access should leave first-year retention largely unaffected even as it affects outcomes concentrated among upperclassmen.

I find that legalized mobile sports betting is associated with a statistically significant 1.21 percentage point decline in four-year graduation rates. This effect grows with each successive cohort’s cumulative exposure to legal betting access. I also find that the gap between female and male six-year graduation rates grows by a marginally significant 0.6 percentage points. This effect does not appear in either the overall six-year graduation rate or first-year retention, both of which are statistically indistinguishable from zero. The 95% confidence interval for first-year retention rules out declines larger than 0.58 percentage points, which is comfortably smaller than the four-year graduation effect, while the six-year graduation rate confidence interval (1.24 percentage points) cannot rule out an effect of similar magnitude to the four-year result. The limited evidence on gender heterogeneity is consistent with a channel concentrated among men. On balance, these results

point to a delay in completion, not a failure to ever complete.

This paper contributes to a growing body of work studying how policy-driven changes in access to amenities or consumption opportunities affect college students. [Blake et al. \(2024\)](#) find that the legalization of recreational marijuana increased college applications, particularly at larger institutions, demonstrating that this type of policy can meaningfully shift student-facing outcomes at the institutional level. [Lindo et al. \(2013\)](#) use within-student variation around their 21st birthdays and find that legal access to alcohol at age 21 lowers students' course grades by roughly 0.03 standard deviations, comparable to a 20-point drop in SAT score. [Braghieri et al. \(2022\)](#) find that the staggered rollout of Facebook across U.S. college campuses worsened student mental health and increased self-reported academic impairment. Together, this literature establishes that access to new amenities, whether a legal substance, a digital platform, or, in this paper's case, a betting product, can have measurable and sometimes unexpected effects on student-related outcomes, motivating closer attention to mobile sports betting specifically.

This paper also contributes to the literature on the financial, psychological, and behavioral consequences of legalized mobile sports betting access, much of it documented amid rising concern about gambling-related harm following *Murphy* ([van der Maas et al., 2022](#); [Yeola et al., 2025](#)). Prior research finds that legalization increases financial distress and reduces household savings and food sufficiency, with effects concentrated among financially constrained households and working-age adults without a college degree ([Hollenbeck et al., 2025](#); [Baker et al., 2024](#); [Guo et al., 2026](#)). Evidence on mental health illustrates the importance of how heterogeneity is measured rather than a genuine disagreement across studies: [Bersak et al. \(2024\)](#) find no effect on self-reported mental health or financial distress across a broad under-40 age bucket, while [Couture et al. \(2024\)](#), using narrower five-year age bins, find that legalization increases the likelihood of poor mental health among men aged 30 to 34 but decreases it among men aged 18 to 24. Other work links legalization to increased substance use and to a reallocation of time away from work and education towards leisure among young adults ([Dasgupta and Ghimire, 2026](#); [Fleming et al., 2024](#)); I build directly on this last finding by testing whether the time reallocated away from education is significant enough to show up in a hard administrative such as graduation. A separate strand documents spillover harms to other household members, including rates of child maltreatment and intimate partner violence ([Wilken, 2024](#); [Arnesen and Matsuzawa, 2025](#)). Despite this growing body of evidence

on the consequences of legalized sports betting, no existing study has yet asked whether these effects extend to whether students actually complete their degree; this paper addresses that gap directly, and does so using an original mobile-specific treatment date rather than the broader and less precise legislative authorization dates used throughout most of this literature.

The remainder of the paper is organized as follows: Section 2 provides information on the background of legalized sports betting in the United States. Sections 3 and 4 outline the data and the empirical strategy, respectively. Section 5 discusses the main results. Section 6 concludes and offers avenues for future research.

2 Background

For over a quarter century, the Professional and Amateur Sports Protection Act of 1992 (PASPA) effectively prohibited sports betting throughout the United States outside of Nevada. New Jersey spent nearly a decade fighting to overturn this restriction, losing repeatedly in federal court before the Supreme Court agreed to hear the case. On May 14, 2018, the Court ruled 6-3 in *Murphy v. NCAA* that PASPA’s prohibition on state authorization of sports betting violated the Tenth Amendment’s anti-commandeering doctrine, striking the law down in its entirety.¹ Then-Governor Chris Christie, who had led New Jersey’s legal fight since 2012, celebrated the ruling as a vindication of states’ rights to regulate gambling within their own borders.² The decision did not legalize sports betting nationwide, however. Instead, it returned the question to the states, and each state was now free to authorize sports wagering on its own timeline.

In the years following *Murphy*, states legalized sports betting at a steady but uneven pace. Some states moved almost immediately: Delaware and New Jersey had mobile betting live within months. Others waited several years, and as of the end of this paper’s sample period in 2024, twenty states had not yet legalized sports betting in any form.

States’ decisions to legalize, and when, were driven primarily by considerations unrelated to higher education, including anticipated tax revenue and political conditions. In fact, [Hollenbeck et al. \(2025\)](#) test this directly using a

¹Murphy v. National Collegiate Athletic Association, 584 U.S. 453 (2018).

²Christie, C. [@GovChristie], quoted in De Vogue, A., & Sneed, T. (2018, May 14). Supreme Court lets states legalize sports gambling. *CNN*. [cnn.com](https://www.cnn.com)

hazard model and find no significant relationship between the timing of state legalization and a range of state-level economic and demographic conditions, including college attainment rates. This staggered, state-by-state rollout is the source of identifying variation in this paper: because the timing of legalization was not driven by conditions at individual colleges or universities, it provides a source of quasi-random variation in students' access to mobile sports betting.

A key institutional detail shapes how this variation should be interpreted: states differ not only in whether they have legalized sports betting, but in whether that betting can be placed online or only in person. Some states permit sports wagering exclusively at licensed retail locations, such as casinos or racetracks, requiring a bettor to physically travel to place a wager. Others additionally permit mobile or online betting, allowing a bettor to wager from a smartphone or computer from anywhere within the state if verified by geolocation software (GeoComply, 2025).³ This distinction is consequential for a population like college students, for whom the marginal cost of placing a bet is plausibly much lower when access requires only a smartphone rather than a trip to a casino. American Gaming Association (2025) document that 95% of the nearly \$150 billion in wagers from Americans in 2024 was bet online rather than at a retail location. Therefore, I define treatment specifically around the availability of legal mobile sports betting, rather than the broader definition of legalization in any form.

Nearly every state that has legalized sports betting restricts wagering to individuals aged 21 or older, consistent with the minimum age typically required for casino gambling. Four of the twenty-nine treated states set the minimum betting age at 18, often because their sports betting programs are administered through state lottery systems rather than traditional casino licensing frameworks.⁴ This near-universal 21-year-old threshold has a direct implication for this paper's empirical design: because most students enter college well before their 21st birthday, the typical entering freshman is not legally permitted to wager when mobile sports betting becomes available in their state. Upperclassmen, particularly juniors and seniors, are far more

³GeoComply handles roughly 95 percent of legal online sports bets placed in North America and utilizes multi-signal methodology (GPS, WiFi, cell-tower triangulation, and IP address plus proprietary databases) as well as anti-spoofing detection for VPNs/proxies/fake-GPS/emulators.

⁴The four states with only an 18+ age restriction are Kentucky, New Hampshire, Rhode Island, and Wyoming. Washington, D.C. is also an 18+ jurisdiction.

likely to have reached the legal betting age while still enrolled. This age structure motivates the use of first-year retention as a falsification check, discussed further below, since a behavioral channel operating through legal access to mobile betting should affect upperclassmen-relevant outcomes well before it affects outcomes concentrated among entering freshmen.

3 Data

The primary data source for this paper is the Integrated Postsecondary Education Data System (IPEDS), a nationwide survey of colleges and universities in the United States. The survey is administered annually by the National Center for Education Statistics (NCES) and is mandatory for any institution participating in Title IV federal financial aid programs, resulting in near-universal coverage of degree-granting institutions in the United States. Institutions report data annually across a series of survey components including student outcomes, providing important measures of student persistence and completion. This paper uses annual IPEDS data spanning 2014 to 2024 on the primary outcomes of interest, four-year and six-year graduation rates. I also examine the first-year retention rate, which serves as a falsification check given that most entering freshmen are below the common legal sports betting age of 21.

The panel is constructed from two IPEDS survey components. The Institutional Characteristics survey (HD) provides the institutional identifiers and characteristics (including control type, level, Carnegie classification, tribal status, and state location) used to apply the sample restrictions described below, while the Graduation Rates survey (GR) provides the outcome variables of interest. The derived IPEDS graduation rate files provide the six-year completion rate disaggregated by sex, though the four-year rate is not provided. Together, these components yield a panel of colleges nested within states, observed annually from 2014 to 2024. After applying the sample restrictions discussed below, the final sample includes 1,488 four-year institutions across 49 states and the District of Columbia. Table 1 presents the full sequence of sample restrictions and their effect on institution count. Because the four-year and six-year graduation rate variables exhibit additional missingness beyond these restrictions, the estimation sample for each outcome is balanced separately. This yields slightly smaller analytic samples of 1,350 institutions for the four-year graduation rate and 1,378 institutions for

the six-year graduation rate.

I also construct an original dataset identifying the date on which each state first offered legal mobile sports betting to the public, distinct from the date a state’s legislature authorized betting in principle. This distinction matters because in some states the date of legislative authorization and the date of practical mobile access diverged significantly, with implications for how treatment should be coded. Iowa represents the most extreme case, where the two dates diverged by over a year and a half.⁵

Treatment dates are constructed from a combination of primary and secondary sources, including legislative documents, state gaming commission records, press releases from sports betting operators and governors’ offices, and local news archives. Where legislative text or commission records were ambiguous about the practical launch of mobile wagering, local news coverage was used to identify the specific date on which the relevant restriction was lifted or the first mobile platform went live. Table 2 reports the date when relevant legislation was signed, the mobile betting launch date, the corresponding first treated academic year, and the source type for each state.

Once a state’s mobile launch date is identified, a second issue remains: because IPEDS measures outcomes at the academic-year level while legalization dates occur on specific calendar days, I adopt a rule to align mobile betting launch dates with academic years. A state’s first treated academic year is the first one whose fall semester begins after the legalization date, using September 30 as a cutoff for whether a launch falls early enough in a given fall to count toward that academic year. Academic years are labeled by their starting calendar year throughout (e.g., the 2021-2022 academic year is

⁵The state’s sports wagering legislation, signed in May 2019, technically permitted mobile and online sports betting from launch. However, the statute required that an account for mobile wagering be established *in person* with a licensed operator: Senate File 617 specifies that “advance deposit sports wagering” accounts must be established by an eligible individual in person with a licensee prior to January 1, 2021 (Iowa General Assembly, 2019). In practice, this meant that an Iowa resident wishing to bet from a smartphone first had to travel to a licensed casino to register before ever placing a bet remotely (Lawhon, 2020). Existing treatment codings, including the widely used American Gaming Association state-of-play map, mark Iowa as having legalized mobile sports betting in 2019, the date the legislation was signed, rather than January 1, 2021, when the in-person registration requirement was lifted and mobile betting became genuinely accessible without ever visiting a casino. For a paper concerned with the behavioral margin of college students, who are unlikely to travel to a licensed casino simply to register for an app, I code Iowa’s treatment date as 2021 rather than 2019.

referred to as 2021). For example, Arizona’s launch on September 9th, 2021 places the law before the cutoff, so that fall’s academic year is treated, while Arkansas’ launch on March 6th, 2022 falls well after the prior fall’s cutoff, so the academic year beginning that fall is the first treated year instead. In both cases, the first treated academic year is the first one in which mobile sports betting was available for nearly its full two semesters.

Table 3 presents summary statistics by treatment status, calculated using pre-treatment years only so that the comparison reflects baseline differences rather than any effect of legalization itself. Treated and never-treated institutions are roughly similar on observable characteristics, though treated institutions exhibit modestly higher baseline persistence outcomes: the average pre-treatment four-year graduation rate is 45.3 percent among institutions in states that eventually legalize mobile sports betting, compared to 38.7 percent among institutions in never-treated states. A similar pattern holds for retention and six-year graduation rates. Academic preparation, measured by institutional SAT Math percentiles, and the gender composition of male applicants, admits, and enrollees are comparable across groups. Identification in a difference-in-differences design does not require treated and never-treated institutions to share the same baseline level of these outcomes, only that they would have followed parallel trends in the absence of legalization, an assumption I assess directly through the event-study estimates in Section 5.1. The final sample includes 962 institutions in eventually-treated states and 526 institutions in never-treated states.

4 Methodology

I identify the impact of legalized mobile sports betting on graduation rates by comparing colleges in never-treated states to recently treated colleges, using both a standard two-way fixed effects (TWFE) estimator and a heterogeneity-robust difference-in-differences estimator to account for the staggered timing of state-level legalization. The basic model being estimated takes the following form:

$$Y_{ist} = \alpha + \beta \text{Legal}_{st} + \mu_i + \lambda_t + \varepsilon_{ist} \quad (1)$$

where Y_{ist} is the outcome of interest, namely four-year graduation rate, six-year graduation rate, or first-year retention rate for college i in state s in year t . First-year retention rate serves primarily as a falsification check, since

most entering freshmen are not yet of legal age to place bets and should therefore be largely unaffected by mobile betting legalization. Legal_{st} is a binary indicator equal to one in the first year state s offers legal mobile sports betting and all subsequent years. μ_i are college fixed effects absorbing all time-invariant institution-level differences (and implicitly state fixed effects since colleges are nested within states). λ_t are year fixed effects absorbing common shocks. ε_{ist} is the error term clustered at the state level, the unit of treatment assignment. β is the coefficient of interest, interpreted as the average effect of mobile sports betting legalization on the outcome.

In settings with staggered treatment adoption, standard TWFE can yield biased results due to already-treated units serving as controls for later-treated units. Therefore, I also implement the technique of [Callaway and Sant’Anna \(2021\)](#), which estimates cohort-specific average treatment effects on the treated, $\text{ATT}(g,t)$, for each group g of states first legalizing in the same year, evaluated at time t , using only never-treated states as the comparison group. These cohort-time specific estimates are then aggregated into a single summary ATT as well as the dynamic event-study estimates presented below. To assess both pre-trends and the dynamic path of treatment effects, I estimate the following event-study specification:

$$Y_{ist} = \alpha + \sum_{k=-5}^3 \beta_k \mathbf{1}[t - G_s = k] + \mu_i + \lambda_t + \varepsilon_{ist} \quad (2)$$

where Y_{ist} is the outcome of interest as defined above. G_s denotes the year in which state s first offered legal mobile sports betting, and $k = t - G_s$ indexes event time (the number of years before or after legalization). $\mathbf{1}[t - G_s = k]$ is an indicator equal to one when college i is k years away from its state’s legalization year. β_k captures the average treatment effect at event time k , estimated using the [Callaway and Sant’Anna \(2021\)](#) estimator with never-treated states serving as the comparison group. The dynamic estimates β_k carry direct structural content in this setting: because IPEDS graduation rates reflect cohort outcomes, β_0 captures the effect on a cohort that was already in its senior year when legalization occurred, while each subsequent β_k reflects a cohort that was one year earlier in their college career at the time of legalization. The dynamic path of post-treatment estimates should therefore be interpreted as tracing out a dose-response relationship in cumulative exposure length. Pre-treatment estimates ($k < 0$) serve as a test of the parallel trends assumption; coefficients statistically indistinguishable

from zero provide evidence that treated and never-treated colleges followed similar graduation trends prior to legalization. μ_i and λ_t denote college and year fixed effects respectively, and ε_{ist} is the error term with standard errors clustered at the state level.

5 Results

5.1 Main Results

Table 4 presents the main results. Panels A and B report the Callaway-Sant’Anna and standard two-way fixed effects estimates, respectively, and columns (1) and (4) reflect estimates for the overall four-year and six-year graduation rates. The TWFE estimates are negative but not statistically significant at conventional levels. The Callaway-Sant’Anna estimate for four-year graduation rates, which accounts for the staggered timing of legalization across states, is larger in magnitude and reaches statistical significance: legalized mobile sports betting is associated with a 1.21 percentage point decline in the four-year graduation rate at affected institutions ($p=0.042$), a decline of roughly 3.1% relative to the pre-treatment mean four-year graduation rate among never-treated institutions.

The overall six-year graduation rate estimates for both the TWFE and Callaway-Sant’Anna estimates are smaller in magnitude and not statistically significant (-0.561 and -0.434, respectively). At the 95% level, I can rule out a reduction in average six-year graduation rates of more than 1.24 percentage points. The contrast between four-year and six-year outcomes is suggestive: a significant decline in four-year completion paired with a smaller, statistically insignificant six-year effect suggests that legalized mobile sports betting may only delay graduation. Students affected by the policy appear more likely to take longer to complete their degree than to fail to complete it at all.

I also run separate analyses for six-year graduation rates by gender in columns (5) and (6), motivated by survey evidence that sports bettors are disproportionately young and male ([Siena College Research Institute, 2025](#)). The Callaway-Sant’Anna estimate for men is negative, though not statistically significant at conventional levels (-0.596, $p=0.135$), while the estimate for women is essentially zero (0.035, $p=0.925$). For women, I can rule out a reduction in six-year graduation rates of more than 0.71 percentage points. Although neither estimate reaches statistical significance, the point estimates

suggest (directionally) a behavioral channel concentrated among male students, the population most likely to hold an active sports betting account. In addition, the gap in six-year graduation rates between men and women grows by 0.6 percentage points, representing about 7.1% of the baseline gender gap among never-treated institutions, though this estimate is only marginally significant ($p=0.064$).

Figure 1 plots the dynamic Callaway-Sant’Anna estimates for the four-year graduation rate. Pre-treatment coefficients are small and statistically indistinguishable from zero. The one exception is $k=-3$, which has a coefficient of -0.460 and is just barely statistically significant at the 5% level ($p=0.043$), supporting the parallel trends assumption underlying the study design. Post-treatment estimates grow steadily more negative: the effect is smallest in magnitude at $k=0$ (-0.51 percentage points) and grows to -1.18 at $k=1$, -1.52 at $k=2$, and -2.78 at $k=3$. As discussed in Section 3, this follows directly from how IPEDS measures graduation cohorts. The cohort reflected in the four-year graduation rate at $k=0$ entered college four years prior to legalization, and so was already well into their college career, and in many cases already in their final year, by the time mobile betting became legal. Each subsequent post-treatment estimate reflects a cohort that entered one year later, and was therefore correspondingly earlier in their college career at the time of legalization, accumulating more potential exposure prior to their expected graduation date. The growing magnitude of the post-treatment estimates traces a dose-response relationship in cumulative exposure to legal mobile sports betting, rather than a one-time shock that dissipates or remains constant over time.

Figure 2 plots the corresponding dynamic estimates for the six-year graduation rate. As with the four-year rate, pre-treatment coefficients are small and statistically indistinguishable from zero at every lead, supporting the parallel trends assumption. Unlike the four-year rate, however, the post-treatment estimates do not exhibit a clear growing pattern and remain statistically insignificant at every horizon, ranging from -0.27 percentage points at $k=0$ to -0.64 at $k=3$. The absence of a significant or clearly widening effect on the six-year graduation rate, set against the significant and growing effect on the four-year rate documented in Figure 1, reinforces the interpretation that legalized mobile sports betting may primarily delay rather than prevent graduation.

Turning back to the gender analysis, Figure 3 plots the corresponding dynamic estimates for six-year graduation rates separately for men and women.

Among men, the post-treatment estimates grow more negative over time, reaching -1.13 percentage points at $k=2$ ($p=0.049$), a pattern similar in shape to the overall four-year result in Figure 1. Among women, the post-treatment estimates remain close to zero at every horizon and never approach statistical significance. This divergence in the dynamic pattern reinforces the cross-sectional comparison in Table 4: the effect of legalized mobile sports betting on six-year graduation appears concentrated among male students.

5.2 Falsification Check: First-Year Retention

Because most entering freshmen are below the legal sports betting age of 21, first-year retention should be largely unaffected by mobile sports betting legalization if the documented effects on graduation operate through a behavioral channel requiring legal access. Table 5 reports the corresponding estimates. Both the TWFE and Callaway-Sant’Anna estimates are small in magnitude and statistically indistinguishable from zero (-0.307 and -0.036, respectively). Figure 5 plots the dynamic estimates, which similarly remain close to zero and statistically insignificant at every post-treatment horizon, with no evidence of the growing pattern documented for the four-year graduation rate. In line with what would be expected under a behavioral channel operating primarily among students old enough to legally wager, I find no significant effect on first-year retention, in contrast to the significant and growing declines in four-year graduation rates documented in Section 5.1.

5.3 Robustness: Student Composition

A potential concern is that the main results could be driven by changes in the composition of students applying to, admitted to, or enrolling at colleges in treated states, rather than by changes in persistence among already-enrolled students. If legalization affected institutions’ applicant pools, for example by deterring risk-averse students or their families, this could generate spurious changes in graduation rates unrelated to any behavioral effect of betting access on enrolled students. Table 6 reports estimates for five measures of student composition: male applicants, male admissions, male first-time enrollment, and institutional SAT Math 25th and 75th percentiles. None of the ten coefficients (five outcomes, two estimators each) is statistically significant at conventional levels, and several point estimates run in the opposite direction from what would be expected if student composition were

changing as a result of legalization. Figures 6-10 plot the dynamic estimates and none suggest an effect, reinforcing the idea that student composition remains unchanged as a result of legalized sports betting. The SAT-based checks warrant some caution, however: many institutions in the sample have adopted test-optional admissions policies over the panel period, and the institutions that continue to report SAT scores, reflected in the smaller sample of 539 institutions for this outcome, are a less representative subset of the full sample than those used in the applicant, admissions, and enrollment checks. Table 7 highlights the differences between reporters and non-reporters, where it is clear that institutions with full SAT data from 2014-2024 are significantly larger and more successful on persistence outcomes than non-reporters.

6 Conclusion

This paper examines whether the legalization of mobile sports betting affects college student persistence, using an original dataset identifying the precise date each state first offered legal mobile wagering combined with an annual panel of 1,488 four-year institutions from 2014 to 2024. Using a heterogeneity-robust difference-in-difference estimator, I find that legalized mobile sports betting is associated with a statistically significant 1.21 percentage point decline in four-year graduation rates, an effect that grows with each successive cohort’s cumulative exposure to legal betting access. The limited evidence on gender heterogeneity is directionally consistent with an effect concentrated among men, though the subgroup estimates do not reach conventional significance. However, the estimate of a growing gap (0.6 percentage points) in six-year graduation rates between women and men is marginally significant. There is no measurable effect for either first-year retention or the overall six-year graduation rate, both of which are statistically indistinguishable from zero.

The contrast between the four-year and six-year results suggests that legalized mobile sports betting primarily delays rather than permanently prevents graduation. The growing magnitude of the effect across successive cohorts, who accumulated progressively more exposure to legal mobile betting over the course of their college career, points in the same direction, reinforcing a cumulative rather than one-time effect. The suggestive concentration of the effect among men, who are also disproportionately likely to hold an active sports betting account, offers some additional support for a

behavioral channel operating through betting access itself, rather than some broader and less targeted state-level shock coinciding with legalization.

The results of this paper contribute to two related literatures. First, it adds to a growing body of work documenting the financial and behavioral consequences of legalized sports betting access, extending this evidence to an educational outcome that has not, to my knowledge, been previously studied in this context. Second, it contributes methodologically by constructing an original dataset that identifies the precise date on which each state first offered legal mobile sports betting, distinct from the date of legislative authorization commonly used in existing research. The Iowa example discussed in Section 3, where a two-year gap separated legislative signing from the lifting of an in-person registration requirement, illustrates why this distinction matters and suggests that studies relying on legislative signing dates alone may meaningfully misclassify treatment timing in settings where the population of interest is particularly sensitive to the convenience of mobile access to sports betting.

This analysis has several limitations. First, IPEDS does not provide individual-level data on student betting behavior, financial distress, or time use, so the mechanism underlying the documented decline in graduation rates cannot be directly observed. The interpretation offered above is in line with the evidence but remains indirect. Second, the derived IPEDS graduation rate files used in this paper provide the six-year completion rate disaggregated by sex, but not the four-year rate, since NCES has historically built out demographic disaggregates around the 150%-of-time completion metric specifically.

Several avenues remain for future work. Most importantly, this paper’s interpretation of the mechanism behind the documented effect remains indirect. Linking this analysis to student-level survey data, such as the National College Health Assessment, would allow future work to test directly whether the decline in four-year graduation rates operates through time reallocation away from coursework, financial distress, or mental health channels associated with problem gambling, rather than inferring mechanism solely from the pattern of results presented here.

From a policy perspective, the findings suggest that the rapid expansion of mobile sports betting access carries consequences for at least one important population beyond the bettors themselves: the institutions and policy-makers responsible for student success. States and universities with large student populations may wish to consider targeted consumer protections or

financial literacy initiatives specifically aimed at college-aged bettors, particularly given that nearly all jurisdictions already restrict betting based on age but few, if any, address the unique circumstances of the college environment.

7 Tables

Table 1: Sample Restrictions

Restriction	Institutions
IPEDS institutions reporting in all years, 2014–2024	5,274
Four-year institutions only	2,388
Drop for-profit institutions	2,165
Drop faith-related special focus institutions	1,900
Primarily baccalaureate-or-above	1,535
Drop tribal colleges	1,534
Drop territories and Nevada	1,488
Institutions Remaining	1,488

Notes: Table presents the sequential application of sample restrictions to the IPEDS panel. Each row reflects the number of unique institutions remaining after applying that restriction in addition to all restrictions above it. Because the four-year and six-year graduation rate variables exhibit additional missingness, the estimation samples for those outcomes are balanced separately and are slightly smaller; see Table 4 for exact sample sizes by outcome.

Table 2: Legal Mobile Sports Betting Launch Dates by State

State	Legislation Signed	Mobile Launch Date	First Treated Academic Year	Source Type
Arizona	April 15, 2021	September 9, 2021	2021	Local news
Arkansas	December 30, 2021	March 6, 2022	2022	Local news
Colorado	May 29, 2019	May 1, 2020	2020	Local news
Connecticut	May 27, 2021	October 19, 2021	2022	Local news
District of Columbia	January 23, 2019	May 28, 2020	2020	Local news
Delaware	PASPA exemption ^b	December 27, 2023	2024	Industry news
Florida	May 25, 2021	December 7, 2023	2024	Local news
Illinois	June 28, 2019	June 18, 2020	2020	Local news
Indiana	May 8, 2019	October 3, 2019	2020	Local news
Iowa	May 13, 2019	January 1, 2021 ^a	2021	Local news
Kansas	May 12, 2022	September 1, 2022	2022	Industry news
Kentucky	March 31, 2023	September 28, 2023	2023	Industry news
Louisiana	June 2021	January 15, 2022	2022	Local news
Maine	May 2, 2022	November 3, 2023	2024	Local news
Maryland	May 18, 2021	November 23, 2022	2023	Local news
Massachusetts	August 10, 2022	March 10, 2023	2023	Local news
Michigan	December 20, 2019	January 22, 2021	2021	Local news
New Hampshire	July 12, 2019	December 30, 2019	2020	Local news
New Jersey	June 11, 2018	June 14, 2018	2018	Local news
New York	April 19, 2021	January 8, 2022	2022	Industry news
North Carolina	June 14, 2023	March 11, 2024	2024	State government
Ohio	December 22, 2021	January 1, 2023	2023	State government
Oregon	PASPA exemption ^b	October 16, 2019	2020	Local news
Pennsylvania	October 30, 2017	May 28, 2019	2019	Industry news
Rhode Island	March 25, 2019	September 4, 2019	2019	Local news
Tennessee	July 1, 2019	November 1, 2020	2021	Local news
Vermont	June 20, 2023	January 11, 2024	2024	Local news
Virginia	April 29, 2020	January 21, 2021	2021	Local news
West Virginia	March 9, 2018	December 27, 2018	2019	Local news
Wyoming	April 5, 2021	September 1, 2021	2021	Industry news

Notes: This table reports the date on which each state’s legislature/governor signed legislation authorizing mobile sports betting, the date on which each state first offered legal mobile sports betting to the public, and the corresponding first treated academic year under the alignment rule described in Section 3. Missouri legalized mobile sports betting on December 1, 2025, after the end of the sample period, and is coded as never-treated for estimation purposes. States not listed had not legalized mobile sports betting as of the end of 2024 and are coded as never-treated. Full source URLs for each state are available upon request.

^a Iowa’s sports wagering legislation was signed in May 2019, but the statute required in-person registration with a licensed operator until January 1, 2021; see Section 3 for discussion.

^b Delaware and Oregon offered sports betting under a pre-existing PASPA exemption rather than new enabling legislation, so no legislation-signing date applies.

Table 3: Summary Statistics by Treatment Status (2014-2017)

	Treated	Control	Difference
<i>Panel A: Persistence outcomes</i>			
Full-time retention rate	77.3 (11.4)	75.0 (11.5)	2.3*** (0.64)
6-year graduation rate	57.8 (18.3)	53.0 (18.2)	4.8*** (1.01)
6-year graduation rate (men)	54.3 (19.3)	49.0 (19.5)	5.2*** (1.10)
6-year graduation rate (women)	61.1 (17.9)	56.5 (17.7)	4.6*** (0.99)
4-year graduation rate	43.8 (21.4)	35.5 (20.3)	8.3*** (1.16)
<i>Panel B: Composition & admissions</i>			
Percent female enrollment	57 (13)	58 (12)	-1 (0.67)
Male applicants	3,051 (4,118)	3,157 (5,761)	-106 (301)
Male admissions	1,686 (2,073)	1,647 (2,235)	40 (126)
Male first-time enrollment	470 (620)	525 (701)	-55 (39)
<i>Panel C: Academic preparation</i>			
SAT Math 25th percentile	491 (75)	484 (68)	7 (4.34)
SAT Math 75th percentile	596 (74)	593 (70)	3 (4.40)
Institutions	962	526	

Notes: This table reports means by treatment status for 2014–2017, the period preceding the May 2018 *Murphy v. NCAA* decision, so that all institutions are observed over the same fixed pre-period regardless of eventual treatment timing. Standard deviations are reported in parentheses below the Treated and Control estimates; the Difference row reports the treated-minus-control gap, with its standard error in parentheses below. Significance stars on the Difference column are from a two-sample t-test allowing unequal variances (Welch). All institution-level outcomes are collapsed to the unit level using 2014–2017 means. SAT scores refer to institutional SAT Math percentiles for admitted students. Observations are at the institution level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects of Legal Access to Mobile Sports Betting on Graduation

	Four-Year Grad Rate			Six-Year Grad Rate			Six-Year Gap
	Overall (1)	Men (2)	Women (3)	Overall (4)	Men (5)	Women (6)	Women—Men (7)
<i>Panel A: Callaway-Sant’Anna</i>							
ATT	-1.206** (0.593)	—	—	-0.434 (0.413)	-0.596 (0.398)	0.035 (0.379)	0.598* (0.323)
Observations	14,850	—	—	15,158	14,575	15,015	14,498
Institutions	1,350	—	—	1,378	1,325	1,365	1,318
Dep. var. mean	44.070	—	—	57.831	53.645	61.541	7.968
<i>Panel B: TWFE</i>							
ATT	-0.868 (0.548)	—	—	-0.561 (0.434)	-0.581 (0.422)	-0.330 (0.416)	0.145 (0.292)
Observations	14,850	—	—	15,158	14,575	15,015	14,498
Institutions	1,350	—	—	1,378	1,325	1,365	1,318
Dep. var. mean	44.070	—	—	57.831	53.645	61.541	7.968

Notes: Callaway-Sant’Anna estimates from [Callaway and Sant’Anna \(2021\)](#), aggregated to a single overall ATT, with never-treated states as the comparison group; TWFE estimates from Equation 1. The outcome of interest is either the four- or six-year graduation rate, measured on a scale of 0 to 100, or, in column (7), the within-institution gap between women’s and men’s six-year graduation rates. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when graduation rates were calculated. Standard errors clustered at the state level are shown in parentheses. Observations are institution-year; institution counts reflect the outcome-specific balanced sample for each column, which is smaller than the full sample of 1,488 institutions reported in Table 1 due to additional missingness in the graduation rate variables. The balanced sample for column (7) requires both male and female graduation rates to be non-missing in every year, so it is smaller than either individual gender sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Falsification Check: First-Year Retention

	TWFE	Callaway-Sant’Anna
ATT	-0.307 (0.290)	-0.036 (0.279)
Institutions	1,381	1,381

Notes: TWFE estimates from Equation 1; Callaway-Sant’Anna estimates from Callaway and Sant’Anna (2021), aggregated to a single overall ATT, with never-treated states as the comparison group. The outcome of interest is first-year retention, measured from 0 to 100 and defined as the percentage of first-year students who return to the same college or university for their second year. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when retention rates were calculated. Standard errors clustered at the state level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Robustness: Student Composition

	TWFE	Callaway-Sant’Anna
Male applicants	-139.97 (152.92)	-47.30 (121.30)
Male admissions	-27.39 (88.95)	-6.58 (70.32)
Male first-time enrollment	-9.21 (11.19)	-10.88 (8.49)
SAT Math 25th percentile	-0.22 (4.49)	1.70 (4.13)
SAT Math 75th percentile	1.18 (3.21)	3.16 (2.39)

Notes: TWFE estimates from Equation 1; Callaway-Sant’Anna estimates from Callaway and Sant’Anna (2021), aggregated to a single overall ATT, with never-treated states as the comparison group. Standard errors clustered at the state level in parentheses. Institution counts range from 539 (SAT percentiles) to 1,277 (male applicants); the smaller SAT sample reflects the growing prevalence of test-optional admissions policies, which limits this check to a larger and more successful (in terms of persistence) subset of institutions that continue to report scores. No estimate is statistically significant at conventional levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

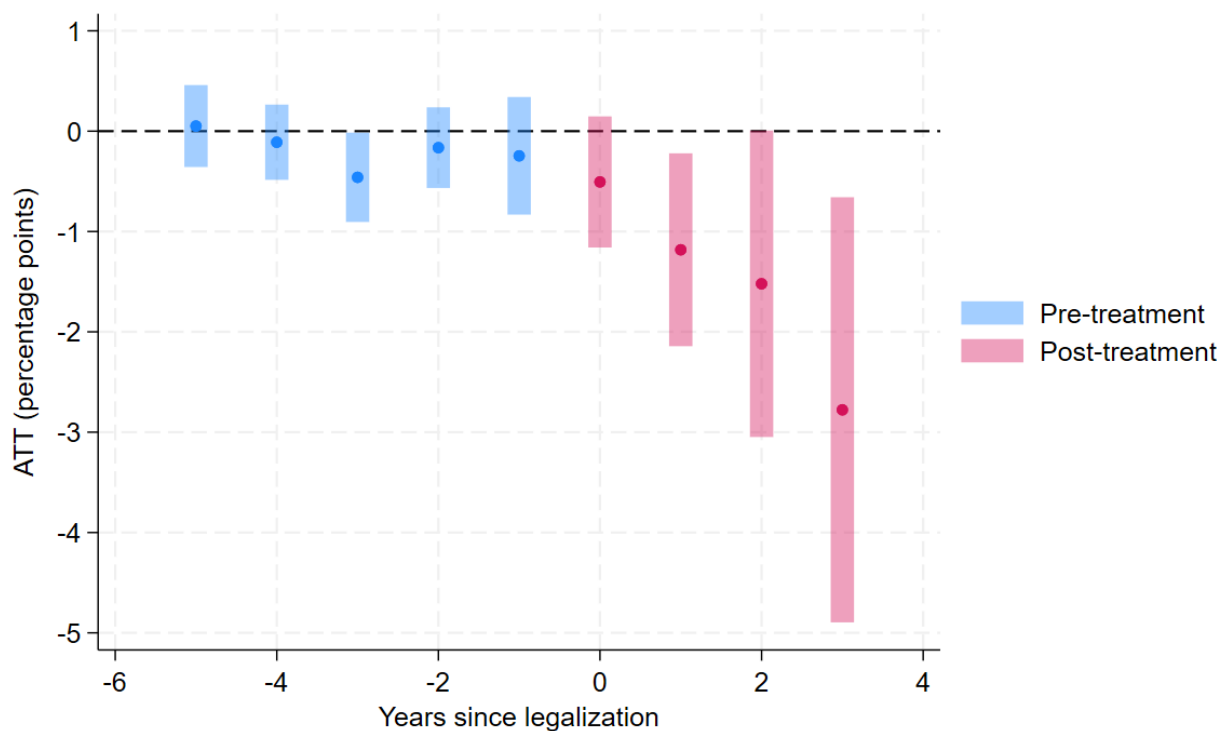
Table 7: Institutional Characteristics by SAT Reporting Status

	Reporters	Non-reporters	Difference
Full-time retention rate	80.5 (9.8)	73.3 (11.0)	7.2*** (0.56)
6-year graduation rate	64.0 (16.9)	53.8 (16.9)	10.2*** (0.92)
4-year graduation rate	49.2 (20.5)	40.3 (18.9)	8.9*** (1.09)
Percent female enrollment	55.9 (11.6)	59.0 (12.6)	-3.2*** (0.65)
Male applicants	4,928 (6,074)	2,579 (5,153)	2,349*** (316)
Male admissions	2,571 (2,863)	1,522 (2,220)	1,049*** (146)
Male first-time enrollment	695 (824)	354 (533)	341*** (40)
Share treated	0.68 (0.47)	0.63 (0.48)	0.05** (0.026)

Notes: This table compares institution-level means between institutions that report SAT Math 25th percentile scores in all 11 sample years (Reporters) and institutions with at least one missing year (Non-reporters), pooling across the full 2014–2024 panel. Standard deviations are reported in parentheses below the Reporters and Non-reporters estimates; the Difference column reports the reporters-minus-non-reporters gap, with its standard error in parentheses below. Institution-level outcomes are collapsed to the unit level using within-institution means. Share treated is the fraction of institutions in each group located in a state that legalized mobile sports betting during the sample period. Significance stars are from a two-sample t-test allowing unequal variances (Welch). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

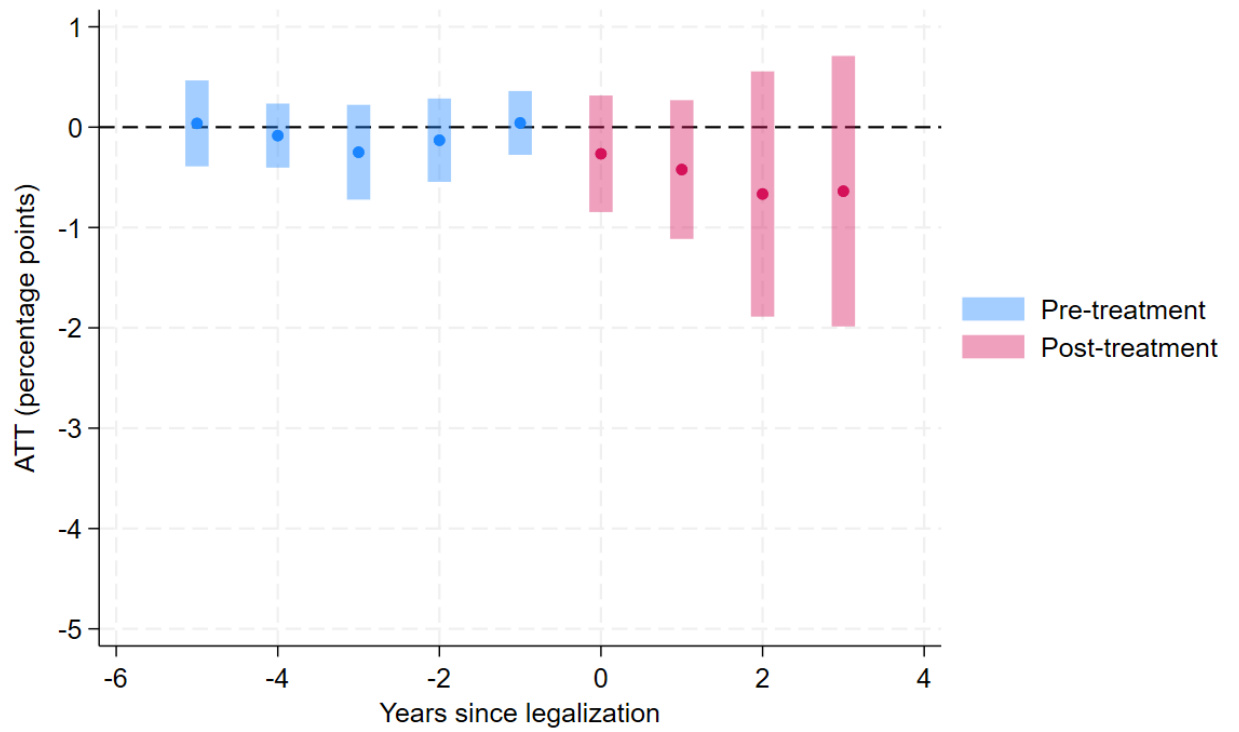
8 Figures

Figure 1: Four-Year Graduation Rate: Dynamic ATT by Event Time



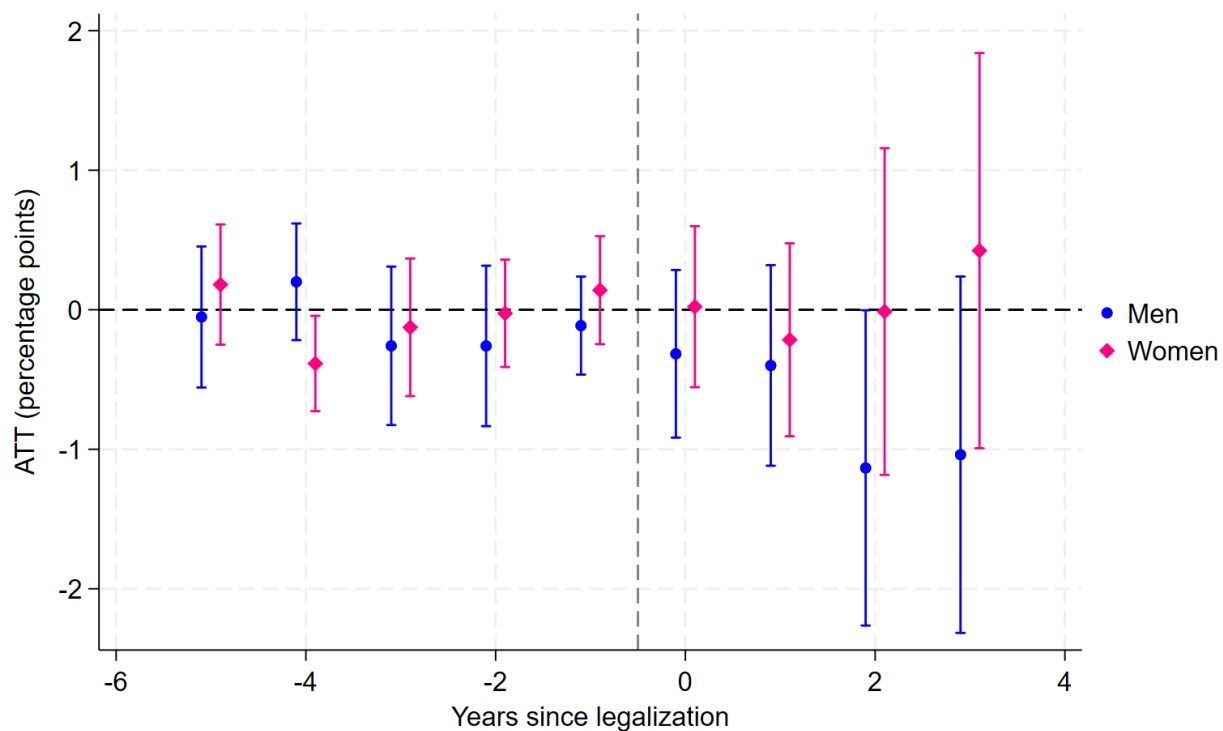
Notes: Estimates from [Callaway and Sant'Anna \(2021\)](#), with never-treated states as the comparison group. The outcome of interest is the four-year graduation rate, measured on a scale of 0 to 100. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when graduation rates were calculated. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 2: Six-Year Graduation Rate: Dynamic ATT by Event Time



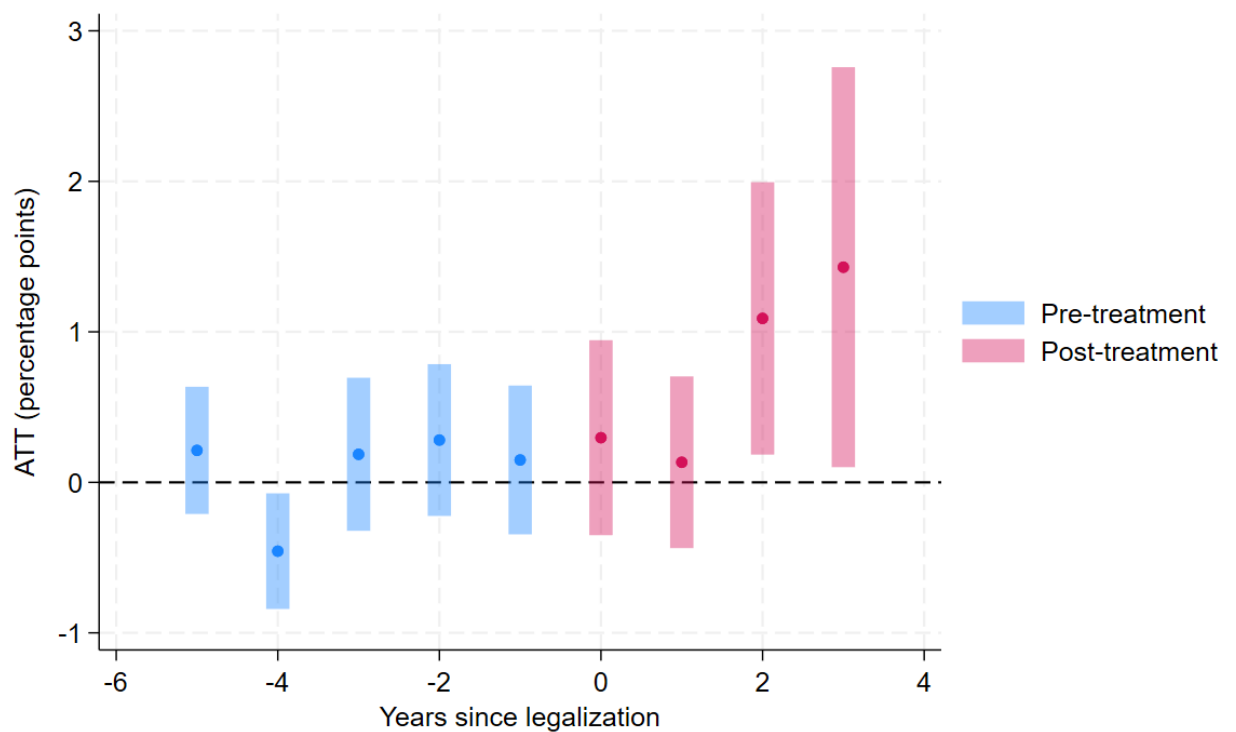
Notes: Estimates from [Callaway and Sant'Anna \(2021\)](#), with never-treated states as the comparison group. The outcome of interest is the six-year graduation rate, measured on a scale of 0 to 100. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when graduation rates were calculated. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 3: Six-Year Graduation Rate by Gender: Dynamic ATT by Event Time



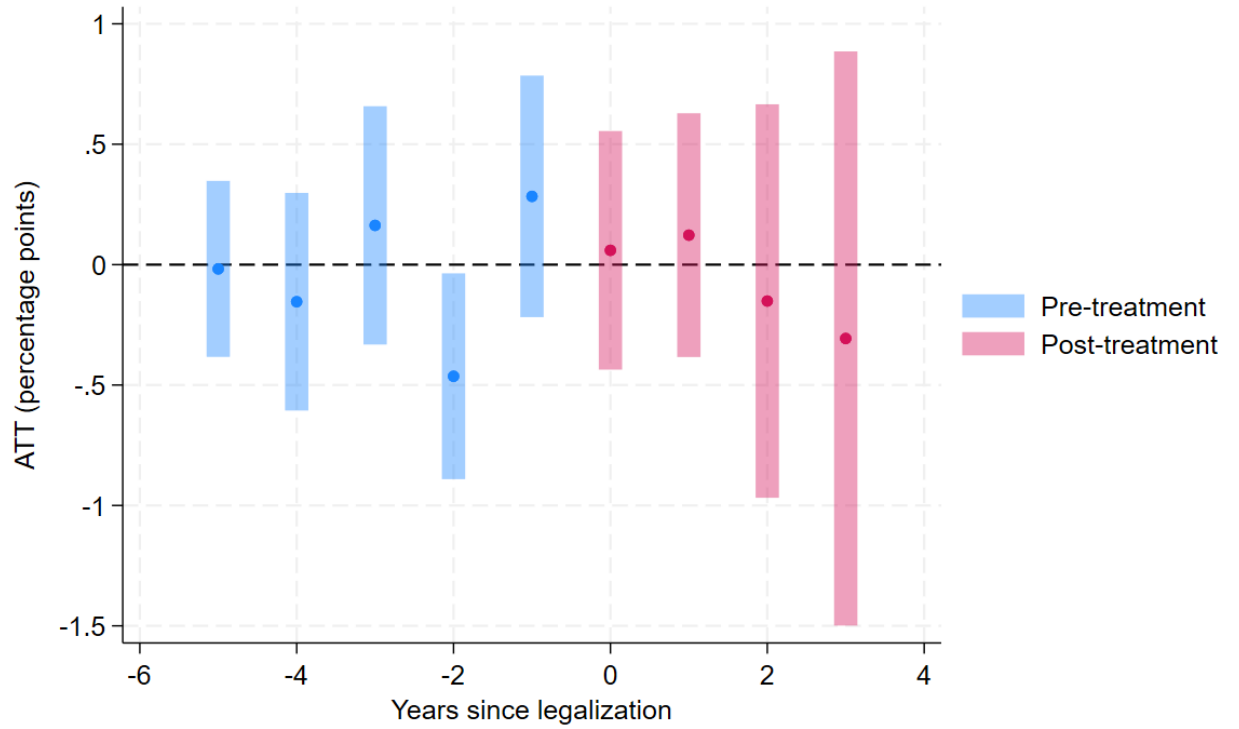
Notes: Estimates from Callaway and Sant'Anna (2021), with never-treated states as the comparison group. The outcome of interest is the six-year graduation rate for men or women, measured on a scale of 0 to 100. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when graduation rates were calculated. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 4: Six-Year Graduation Rate Gap (Women - Men): Dynamic ATT by Event Time



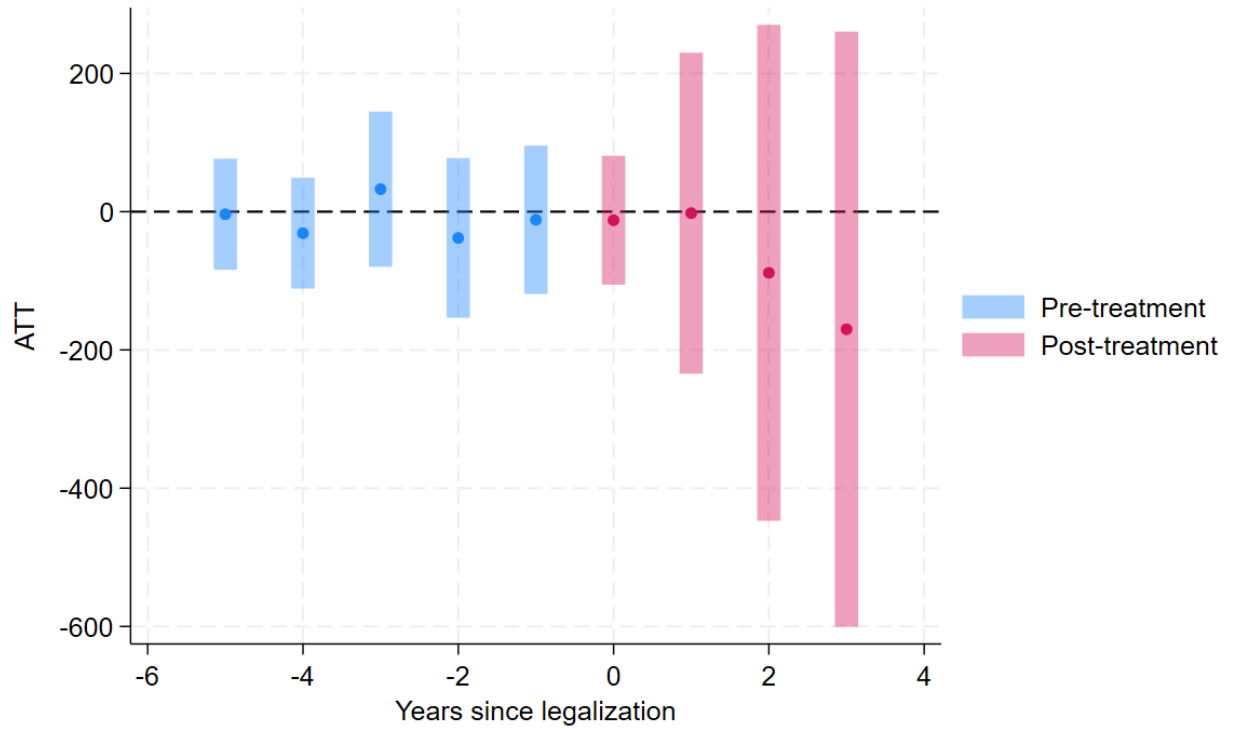
Notes: Estimates from Callaway and Sant'Anna (2021), with never-treated states as the comparison group. The outcome of interest is the within-institution gap between women's and men's six-year graduation rates. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when graduation rates were calculated. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 5: First-Year Retention: Dynamic ATT by Event Time



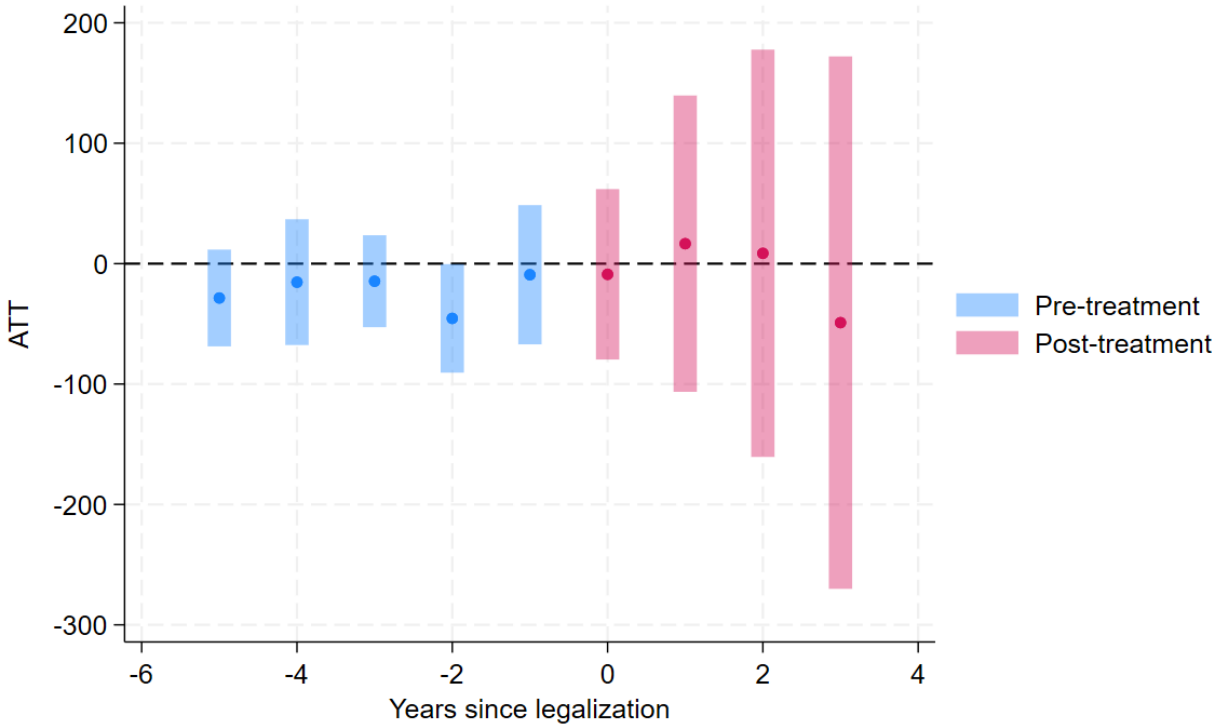
Notes: Estimates from Callaway and Sant'Anna (2021), with never-treated states as the comparison group. The outcome of interest is first-year retention, measured from 0 to 100 and defined as the percentage of first-year students who return to the same college or university for their second year. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when retention rates were calculated. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 6: Composition (Male Applicants): Dynamic ATT by Event Time



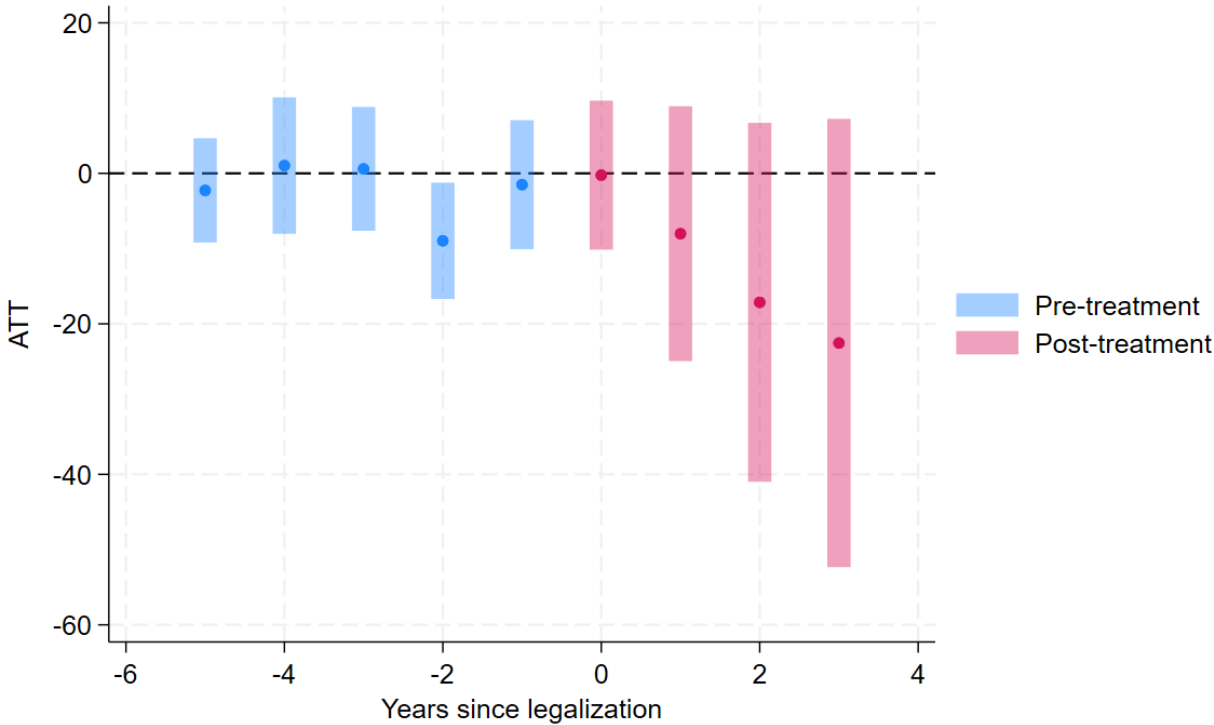
Notes: Estimates from [Callaway and Sant'Anna \(2021\)](#), with never-treated states as the comparison group. The outcome of interest is the number of male applications received. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when applications were received. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 7: Composition (Male Admissions): Dynamic ATT by Event Time



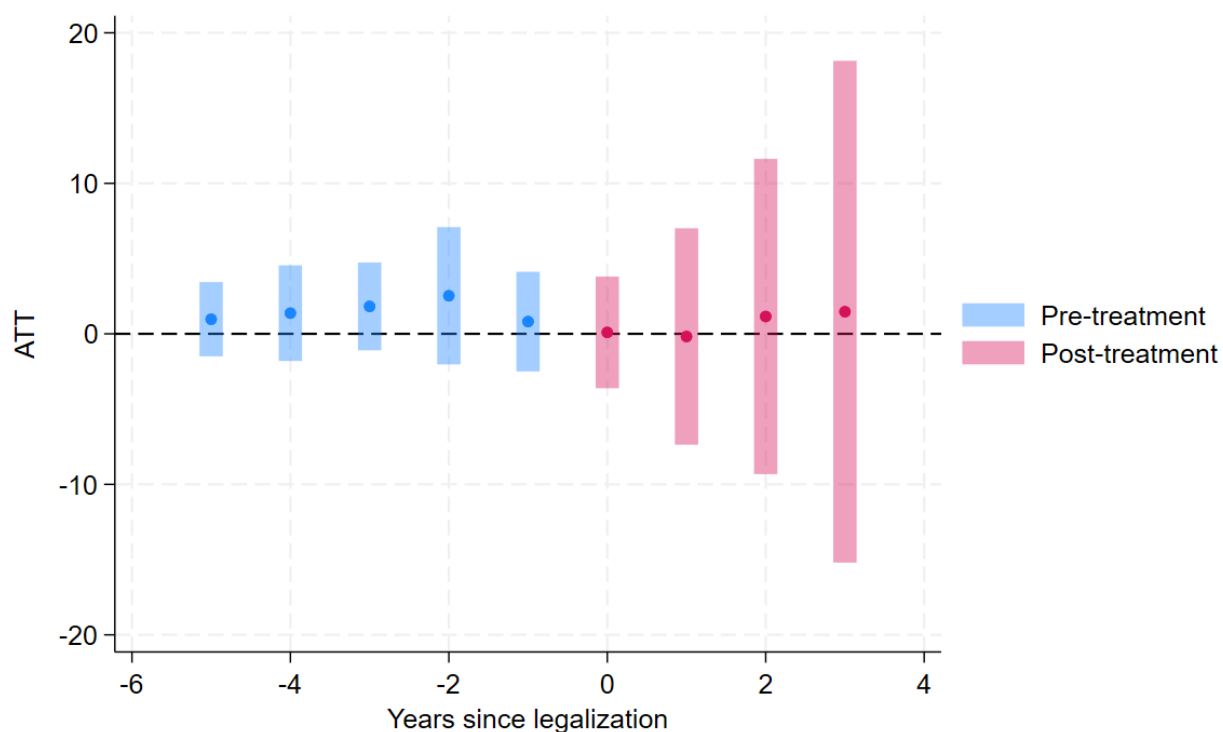
Notes: Estimates from [Callaway and Sant'Anna \(2021\)](#), with never-treated states as the comparison group. The outcome of interest is the number of male students admitted. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when applications were received. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 8: Composition (Male Enrollment): Dynamic ATT by Event Time



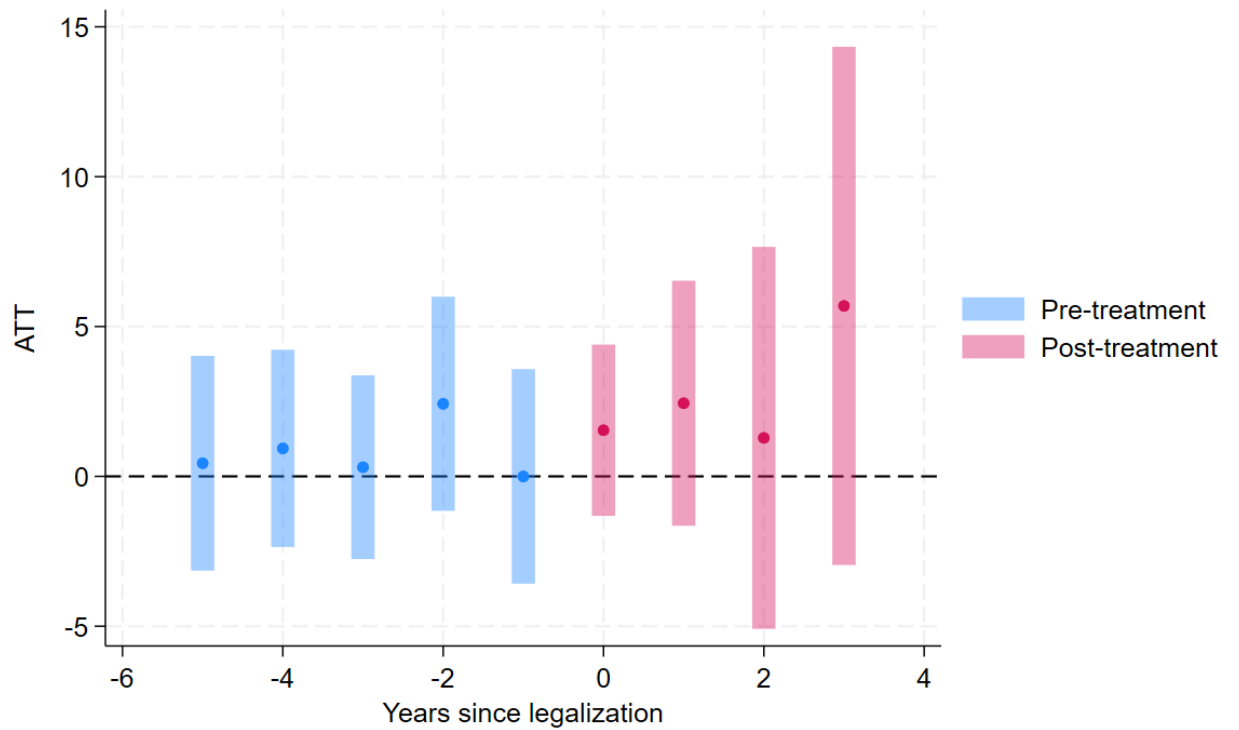
Notes: Estimates from Callaway and Sant’Anna (2021), with never-treated states as the comparison group. The outcome of interest is the number of male students entering postsecondary education for the first time. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when applications were received. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 9: Composition (SAT Math 25th percentile): Dynamic ATT by Event Time



Notes: Estimates from [Callaway and Sant'Anna \(2021\)](#), with never-treated states as the comparison group. The outcome of interest is the average score of the 25th percentile of the math portion of the SAT. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when applications were received. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

Figure 10: Composition (SAT Math 75th percentile): Dynamic ATT by Event Time



Notes: Estimates from Callaway and Sant'Anna (2021), with never-treated states as the comparison group. The outcome of interest is the average score of the 75th percentile of the math portion of the SAT. The treatment variable is a binary indicator for whether the state had legalized mobile sports betting prior to when applications were received. Event-time window restricted to -5 to $+3$ to avoid composition-driven artifacts from cohort thinning at longer horizons. Standard errors clustered at the state level.

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